Efficient Harmonic Balance Analysis of Waveguide Devices with Nonlinear Dielectrics

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***Abstract*—The computation of high order harmonic components of the time-harmonic field originated in waveguide structures containing nonlinear dielectrics is addressed exploiting harmonic balance and finite element methods. The solution of the nonlinear system is handled either via Picard or relaxed iterations. Being the nonlinear loop particularly time demanding, a domain decomposition method is proposed to speed-up the system solution by restricting the iteration to a small subdomain. A numerical example over a 2-D H-plane device is presented.**

***Index Terms*—Harmonic analysis, finite element methods, domain decomposition method, nonlinear media, waveguide filters.**

I. Introduction

TIME-DOMAIN analysis of waveguide devices including nonlinear dielectrics often results to be computationally prohibitive [1]. A time-harmonic finite element (FE) approach at a single frequency is not sufficient to include all relevant phenomena. Higher order harmonics are needed and may be taken into account by using the harmonic balance finite element (HBFE) method [2], also known as multiharmonic finite element method [3], which provides a fast and accurate solution for low frequency problems. In this framework, an algebraic domain decomposition (DD) technique was proposed to improve computational timings also in high frequency problems [4], but no harmonic balance was present there and hence only the fundamental frequency was evaluated. In this letter, [4] is expanded by implementing a full harmonic balance domain decomposition finite element (HBDDFE) method for solving Helmholtz equation. The capability and efficiency of the method are proved on an H-plane waveguide filter example.

II. Formulation

Given a generic multiport device, its domain Ω either in 2-D or 3-D Euclidean space can be divided into a large part Ω_1 comprising only linear media, and a small part Ω_2 comprising all nonlinear media. Permittivity in Ω can be expressed as

$$\varepsilon_r(\mathbf{E}(\mathbf{r}),\mathbf{r})=\begin{cases}\bar{\varepsilon}_r(\mathbf{r}), & \mathbf{r}\in\Omega_1\\ \tilde{\varepsilon}_r(\mathbf{E}(\mathbf{r}),\mathbf{r})=\bar{\varepsilon}_r(\mathbf{r})+\mathcal{L}(\mathbf{E}(\mathbf{r})), & \mathbf{r}\in\Omega_2\end{cases} \quad (1)$$

where $\mathbf{E}(\mathbf{r})$ is the electric field in the generic point $\mathbf{r}$ and $\mathcal{L}(\cdot)$ denotes a generic scalar operator describing the nonlinear behavior [4]. The electric field satisfies, within Ω, the well known vector Helmholtz equation with suitable boundary conditions on Γ. For the example shown in Section III (Fig. 1), Γ comprises a part Γ_0 which will be considered perfect electric conductor, and two parts Γ_1 and Γ_2 corresponding to the ports where a modal expansion is enforced [5]. Γ_{12} corresponds to the internal boundary between subdomains Ω_1 and Ω_2.

In a standard FE Galerkin framework the domain Ω is discretized into a set of non overlapping finite elements and the electric field is approximated, in the 3-D case, as a linear combination of vector (edge) basis functions $\boldsymbol{\alpha}_n$. In the H-plane (2-D) case treated in Section III, scalar nodal basis functions multiplyed by the out-of-plane unit vector (TM problem) are employed. By testing Helmholtz equation over Ω with weighting functions $\boldsymbol{\alpha}_m$, chosen equal to the basis, the following weak form is obtained:

$$\int_\Omega \nabla\times\boldsymbol{\alpha}_m(\mathbf{r})\cdot\frac{1}{\mu_r}\nabla\times\mathbf{E}(\mathbf{r})\;d\Omega-k^2\int_\Omega\boldsymbol{\alpha}_m(\mathbf{r})\cdot\varepsilon_r(\mathbf{E}(\mathbf{r}),\mathbf{r})\,\mathbf{E}(\mathbf{r})\;d\Omega+\oint_\Gamma\boldsymbol{\alpha}_m(\mathbf{r})\cdot\hat{n}\times\frac{1}{\mu_r}\nabla\times\mathbf{E}(\mathbf{r})\;d\Gamma=0 \quad (2)$$

being k the free space wavenumber.

The multiharmonic dependence in the HBFE method is introduced by approximating the electric field as truncated Fourier series [3]

$$\hat{\mathbf{E}}(\mathbf{r},t)=\sum_{p=1}^{P}\left(\mathbf{E}_p^{(s)}(\mathbf{r})\sin(p\omega_0t)+\mathbf{E}_p^{(c)}(\mathbf{r})\cos(p\omega_0t)\right) \quad (3)$$

where ω_0 is the fundamental angular frequency and $\mathbf{E}_p^{(s)}$ and $\mathbf{E}_p^{(c)}$ are the harmonic complex amplitudes that expand the field at $p\omega_0$ and hence they must individually satisfy Helmholtz equation with $k=pk_0$. This is done by using the FE expansion for each coefficient $\mathbf{E}_p^{(s)}$ and $\mathbf{E}_p^{(c)}$. The scalar relative permittivity can also be approximated as [3]

$$\hat{\varepsilon}_r(\mathbf{E}(\mathbf{r}),\mathbf{r},t)=\varepsilon_{r_0}+\sum_{g=1}^{G}\left(\varepsilon_{r_g}^{(s)}\sin(g\omega t)+\varepsilon_{r_g}^{(c)}\cos(g\omega t)\right) \quad (4)$$

where the coefficients of the expansion are given by

$$\begin{aligned}\varepsilon_{r_0}&=\frac{\omega}{2\pi}\int_0^{\frac{2\pi}{\omega}}\varepsilon_r(\mathbf{E}(\mathbf{r}),\mathbf{r})\;dt\\ \varepsilon_{r_g}^{(s)}&=\frac{\omega}{\pi}\int_0^{\frac{2\pi}{\omega}}\varepsilon_r(\mathbf{E}(\mathbf{r}),\mathbf{r})\sin(g\omega t)\;dt\\ \varepsilon_{r_g}^{(c)}&=\frac{\omega}{\pi}\int_0^{\frac{2\pi}{\omega}}\varepsilon_r(\mathbf{E}(\mathbf{r}),\mathbf{r})\cos(g\omega t)\;dt\end{aligned} \quad (5)$$

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A conventional, linear, FE approach discretization of (2) leads to a system in the form $[a_{mn}][E_n] = [b_m]$ [5]. In an HBFE approach, $\mathbf{E}(\mathbf{r})$ and $\varepsilon_r(\mathbf{r})$ must be substituted by their respective harmonic approximations $\hat{\mathbf{E}}(\mathbf{r}, t)$ and $\hat{\varepsilon}_r(\mathbf{r}, t)$ in (2). By performing a further testing with weights $\sin(q\omega t)$ or $\cos(q\omega t)$, $q = 1 \ldots P$, and integrating over the time period of the fundamental $\left[0, \frac{2\pi}{\omega}\right]$, a large linear system is obtained which has formally the same structure as a conventional FE system but in which each entry a_{mn}, E_n or b_m is itself a matrix or a vector:

$$a_{mn} = \begin{bmatrix} a_{mn}^{(1s1s)} & a_{mn}^{(1s1c)} & \ldots & a_{mn}^{(1sPc)} \\ a_{mn}^{(1c1s)} & a_{mn}^{(1c1c)} & \ldots & a_{mn}^{(1cPc)} \\ \vdots & \vdots & \ddots & \vdots \\ a_{mn}^{(Pc1s)} & a_{mn}^{(Pc1c)} & \ldots & a_{mn}^{(PcPc)} \end{bmatrix} \quad (6)$$

$$E_n = \begin{bmatrix} E_n^{(1s)} & E_n^{(1c)} & \ldots & E_n^{(Pc)} \end{bmatrix}^{\mathrm{T}} \quad (7)$$

$$b_m = \begin{bmatrix} b_m^{(1s)} & b_m^{(1c)} & \ldots & b_m^{(Pc)} \end{bmatrix}^{\mathrm{T}} \quad (8)$$

The superscripts $(q[s|c])$ and $(p[s|c])$ indicates that the corresponding entry was computed by testing (2) with $[\sin(q\omega t)|\cos(q\omega t)]$ while the corresponding harmonic basis in (3) was $[\sin(p\omega t)|\cos(p\omega t)]$.

The nonlinearity is finally handled via a relaxed iteration. As a starting point a null electric field $[E]^0$ is assumed, computing the pertinent system matrix $[A_{([E]^0)}]$ and known term $[b_{([E]^0)}]$ and solving the system $[A_{([E]^0)}][E]^1 = [b_{([E]^0)}]$. The newly computed field $[E]^1$ can now be used to update system matrix and known term and perform the following relaxed iteration:

$$[A_{(\gamma[E]^{i-1}+(1-\gamma)[E]^{i-2})}][E]^i = [b_{(\gamma[E]^{i-1}+(1-\gamma)[E]^{i-2})}] \quad (9)$$

with $i \geq 2$ and $\gamma \in (0, 1]$. $\gamma = 1$ gives the standard Picard iteration [4], while low γ values damps the oscillations which may arise with highly nonlinear materials or high intensity impressed fields at the cost of a slower convergence [6]. The process is repeated until the relative error between the updated solution and the previous one, in the sense of the Euclidean norm, is less than a prescribed value τ [4].

The order of the resulting HBFE system is $2NP$, and this number can become very high, and, since the solutions have to be refined iteratively building at each step a new linearized system, computing time can become too high. To enhance the computational efficiency, a substructuring DD approach is here proposed [4], [7]. The multiharmonic unknowns vector $[E]$ defined by the HBFE method over Ω can be split into two vectors $[E^{(1)}]$ and $[E^{(2)}]$ containing the unknowns belonging to interior points in Ω_1 and Ω_2, respectively. Unknowns belonging to the boundaries $\Gamma \cup \Gamma_{12}$ are placed in a third vector $[E^{(c)}]$. The HBFE system can then be recast in:

$$\begin{bmatrix} [M^{(1)}] & 0 & [P^{(1)}] \\ 0 & [M^{(2)}]^{i-1} & [P^{(2)}]^{i-1} \\ [Q^{(1)}] & [Q^{(2)}]^{i-1} & [C]^{i-1} \end{bmatrix} \begin{bmatrix} [E^{(1)}]^i \\ [E^{(2)}]^i \\ [E^{(c)}]^i \end{bmatrix} = \begin{bmatrix} [b^{(1)}] \\ [b^{(2)}]^{i-1} \\ [b^{(c)}]^{i-1} \end{bmatrix} \quad (10)$$

where $[M^{(1)}]$ contains the HBFE coefficients of the linear system related to the unknowns $[E^{(1)}]$, with null harmonic coupling coefficients, while $[M^{(2)}]$ contains coefficients for the nonlinear subdomain Ω_2. $[P^{(\cdot)}]$ and $[Q^{(\cdot)}]$ represent couplings between interior unknowns and boundary unknowns collected in $[E^{(c)}]$. By using the Schur complement concept, the boundary unknowns $[E^{(c)}]^i$ of (10) and hence the generalized scattering matrix of the device can be retrieved with the same accuracy of the full domain solution [7]. Thus, in order to solve the HBDDFE system, every single submatrix is assembled at first sight, then, within the iteration loop, only the submatrices related to Ω_2 and Γ_{12} are to be updated. The efficiency of this DD technique relies on the restriction of matrices updates to sufficiently small regions along the nonlinear iterative solution, noticeably improving long nonlinear loops.

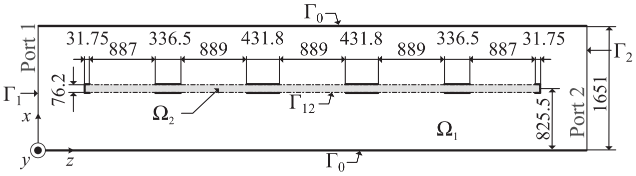

Fig. 1. H-plane cross section of a D-Band passband filter (dielectric slab is shown in light grey; metalization is shown as thick black lines) in a WR6 waveguide. Measures are given in μm.

III. Numerical Results

A millimeter wave passband filter in WR6 (1651x825.5 μm) rectangular waveguide is here analyzed [4]. The filter, uniform along y axis, is realized by placing on the E-plane a dielectric slab partially metalized on both sides as shown by the H-plane cross section of Fig. 1. All conductors are considered perfect. The dielectric, enclosed in Ω_2, presents a Kerr-type nonlinearity of such that [4]

$$\tilde{\varepsilon}_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) = \bar{\varepsilon}_r(\mathbf{r}) + \alpha_2\ |\mathbf{E}(\mathbf{r})|^2, \qquad \mathbf{r} \in \Omega_2 \quad (11)$$

where $\bar{\varepsilon}_r(\mathbf{r}) = 2.1$ and $\alpha_2 = 1.625\ 10^{-10}\ \mathrm{m}^2/\mathrm{V}^2$. The Kerr-like behavior of the permittivity induces the generation of odd order harmonics, thus even ones can been neglected. For the relaxed iteration stop criterion, $\tau = 10^{-6}$ has been chosen. The continuity of the field at the ports is imposed through a modal expansion exploiting only TE_{m0} modes, $m = 2n + 1$, $n = 1, \ldots, 4$, even m modes being absent for the E-plane symmetry of the filter. The excitation is a TE_{10} mode at fundamental frequency impinging at Port 1 with maximum field amplitude E_i. The waveguide discontinuity represented by the filter transfers power to higher order modes, which may result to be guided at higher harmonic frequencies. Furthermore, as a sinusoidal excitation is chosen, only $\sin(p\omega t)$ related coefficients are retained. Due to H-plane uniformity, the problem can be treated as 2-D [5]. First order nodal elements are used and the discretization leads to 2473 degrees of freedom, 80 of which belong to the nonlinear subdomain Ω_2, 314 to the boundaries and 2079 to the linear subdomain Ω_1. Such a fine discretization is necessary to obtain a good approximation of the field up to the 5^{th} harmonic.

As a first analysis, the effect of the total number of harmonics P retained on the filter's spectral response is investigated (Fig. 2). For an incident field amplitude $E_i = 10$ kV/m, the

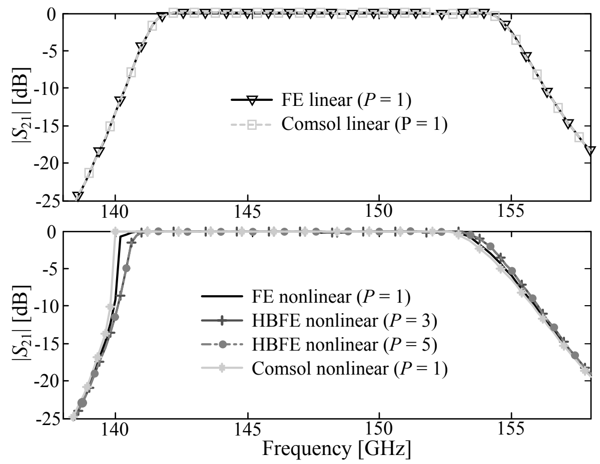

Fig. 2. TE_{10} spectral response for linear (top) and nonlinear (bottom) permittivity with $E_i = 10$ kV/m. The results are compared with Comsol RF module linear and nonlinear solutions.

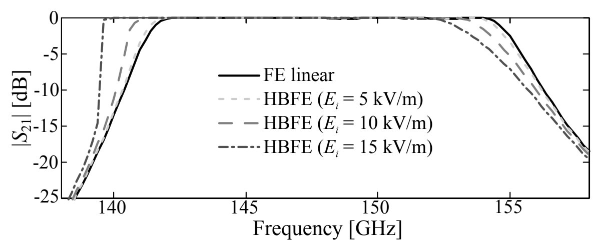

Fig. 3. $|S_{21}|$ of the nonlinear filter for several values of E_i compared to the linear case. Field is approximated up to the 5^{th} harmonic.

Picard iteration proved to converge. Equal mesh and parameters have been used to conduct a 2-D FE (single harmonic) analysis with Comsol RF Module [8]. Being the nonlinear loop tackled differently (Comsol uses Newton algorithm), perfect matching in the spectral response is not obtained as for the linear permittivity case ($\alpha_2 = 0$).

To gain better insight, simulations have been repeated with $E_i = 5$ kV/m and $E_i = 15$ kV/m, for $P = 5$. The results compared to those of a linear device are shown in Fig. 3. There is an evident shift of the band towards lower frequencies as higher power densities are involved, which agrees to [9]. Furthermore the $E_i = 15$ kV/m did not converge with Picard iteration and required a relaxed $\gamma = 0.1$ iteration.

Analysis of the spectral response convergence is performed computing the relative error, in Euclidean norm sense, all over the device bandwidth while increasing the order of HBFE system such that

$$\mathrm{error}(P = p) = \frac{\| |S_{21}|^p - |S_{21}|^{p-2} \|_2}{\| |S_{21}|^{p-2} \|_2} \tag{12}$$

with $p = 2n + 1$, $n = 1, \ldots, 5$. Results (Fig. 4) show faster convergence for low intensity impinging fields.

The efficiency of the DD technique can be assessed by its acceleration, that is the ratio between time for a full domain computation and the corresponding DD computation (comprehensive of the Schur complement solution) [4]. Fig. 5 presents the acceleration for each frequency point, and for different values of P at $E_i = 15$ kV/m. In general, higher

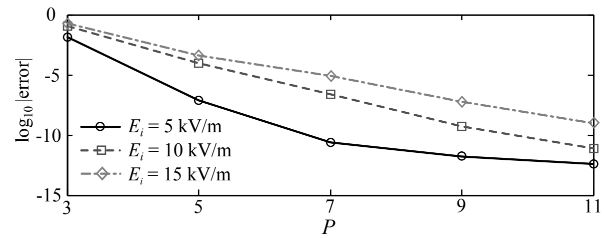

Fig. 4. Relative error of the spectrum at fundamental frequency for various E_i values.

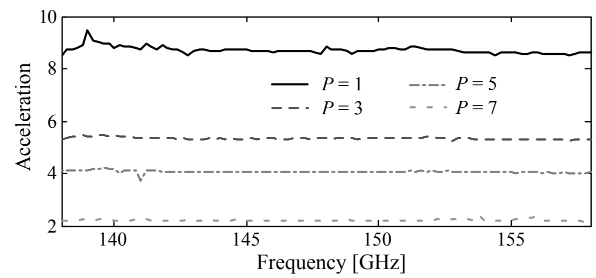

Fig. 5. Acceleration spectrum obtained for an HBFE system of several harmonic orders. The computations are made for $E_i = 15$ kV/m ($\gamma = 0.1$).

order HBDDFE systems have lower acceleration, since the degrees of freedom dimensions related to nonlinear media increase.

IV. Conclusion

A harmonic balance approach to the DD-FE solution for the efficient analysis of waveguide devices containing nonlinear dielectrics has been presented and the accuracy of the multiharmonic solution discussed.

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